

torch.sparse_bsc_tensor#

https://pytorch.org/docs/stable/generated/torch.sparse_bsc_tensor.html

Description:

Constructs a sparse tensor in BSC (Block Compressed Sparse Column) with specified 2-dimensional blocks at the given `ccol_indices` and `row_indices`. Sparse matrix multiplication operations in BSC format are typically faster than that for sparse tensors in COO format. Make you have a look at the note on the data type of the indices. If the device argument is not specified the device of the given values and indices tensor(s) must match. If, however, the argument is specified the input Tensors will be converted to the given device and in turn determine the device of the constructed sparse tensor. `ccol_indices` (array_like) – (B+1)-dimensional array of size (*batchsize, ncolblocks + 1). The last element of each batch is the number of non-zeros. This tensor encodes the index in values and `row_indices` depending on where the given column starts. Each successive number in the tensor subtracted by the number before it denotes the number of elements in a given column. `row_indices` (array_like) – Row block coordinates of each block in values. (B+1)-dimensional tensor with the same length as values.

Function Signature

```
torch.sparse_bsc_tensor(ccol_indices, row_indices, values,  
size=None, *, dtype=None, device=None, pin_memory=False,  
requires_grad=False, check_invariants=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

`dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, infers data type from values. `device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. `pin_memory` (bool, optional) – If set, returned tensor would be allocated in the pinned memory. Works only for CPU tensors. Default: False. `requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default: False. `check_invariants` (bool, optional) – If sparse tensor invariants are checked. Default: as returned by `torch.sparse.check_sparse_tensor_invariants.is_enabled`

Code Examples

```
>>> ccol_indices = [0, 1, 2]
>>> row_indices = [0, 1]
>>> values = [[1, 2], [3, 4]], [[5, 6], [7, 8]]
>>> torch.sparse_bsc_tensor(torch.tensor(ccol_indices,
...                                     dtype=torch.int64),
...                           torch.tensor(row_indices,
...                                     dtype=torch.int64),
...                           torch.tensor(values,
...                                     dtype=torch.double)
tensor(ccol_indices=tensor([0, 1, 2]),
      row_indices=tensor([0, 1]),
      values=tensor([1., 2.],
                    [3., 4.],
                    [5., 6.],
                    [7., 8.])), size=(2, 2), nnz=2,
      dtype=torch.float64,
      layout=torch.sparse_bsc)
```

Notes & Warnings

NOTE

Note If the device argument is not specified the device of the given values and indices tensor(s) must match. If, however, the argument is specified the input Tensors will be converted to the given device and in turn determine the device of the constructed sparse tensor.

Detailed Documentation

Documentation

Constructs a sparse tensor in BSC (Block Compressed Sparse Column)) with specified 2-dimensional blocks at the given `ccol_indices` and `row_indices`. Sparse matrix multiplication operations in BSC format are typically faster than that for sparse tensors in COO format. Make you have a look at the note on the data type of the indices.

If the device argument is not specified the device of the given values and indices tensor(s) must match. If, however, the argument is specified the input Tensors will be converted to the given device and in turn determine the device of the constructed sparse tensor.

`ccol_indices` (array_like) – (B+1)-dimensional array of size (*batchsize, ncolblocks + 1). The last element of each batch is the number of non-zeros. This tensor encodes the index in values and `row_indices` depending on where the given column starts. Each successive number in the tensor subtracted by the number before it denotes the number of elements in a given column.

`row_indices` (array_like) – Row block coordinates of each block in values. (B+1)-dimensional tensor with the same length as values.

`values` (array_list) – Initial blocks for the tensor. Can be a list, tuple, NumPy ndarray, and other types that represents a (1 + 2 + K)-dimensional tensor where K is the number of dense dimensions.

`size` (list, tuple, torch.Size, optional) – Size of the sparse tensor: (*batchsize, nrows * blocksize[0], ncols * blocksize[1], *densesize) If not provided, the size will be inferred as the minimum size big enough to hold all non-zero blocks.

`dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None,

infers data type from values.

`device` (`torch.device`, optional) – the desired device of returned tensor. Default: if `None`, uses the current device for the default tensor type (see `torch.set_default_device()`). `device` will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

`pin_memory` (`bool`, optional) – If set, returned tensor would be allocated in the pinned memory. Works only for CPU tensors. Default: `False`.

`requires_grad` (`bool`, optional) – If autograd should record operations on the returned tensor. Default: `False`.

`check_invariants` (`bool`, optional) – If sparse tensor invariants are checked. Default: as returned by `torch.sparse.check_sparse_tensor_invariants.is_enabled()`, initially `False`.